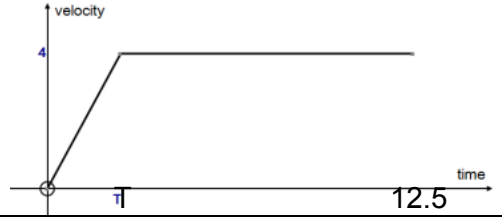
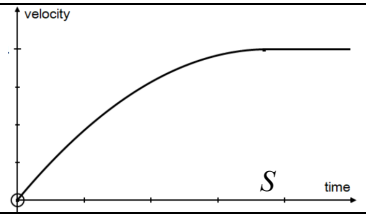
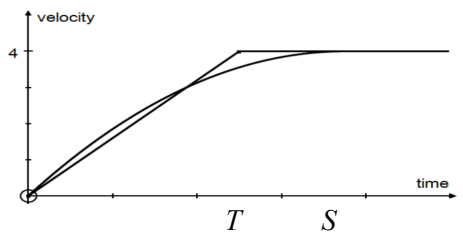
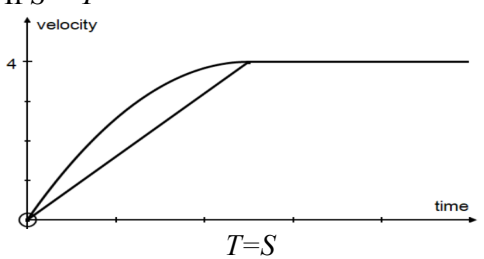
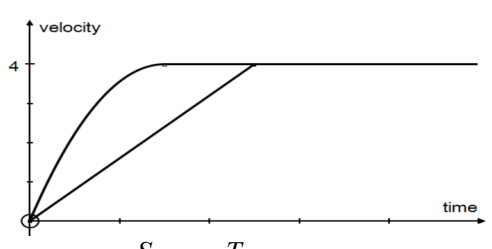
| Question | | | Answer | Marks | AOs | | Guidance |
|---|---|---|---|---|---|---|---|
| **10** | i | |  | **B1**<br><br>**B1**<br><br>**[2]** | **1.1a**<br><br>**1.1a** | Two line segments with one horizontal<br><br>$(T, 4)$ and (12.5, 4) labelled or indicated on scales. Allow their 2.5 marked instead of $T$ .<br>On axes labelled $v$ and $t$ oe | |
| | ii | | $\frac{1}{2}\times 4\times\left(12.5+\left(12.5-T\right)\right)=45$<br><br>$T=2.5$ | **M1**<br><br>**A1**<br>**[2]** | **3.1a**<br><br>**1.1b** | Attempt to find area of trapezium or both the the triangle $\left(\frac{1}{2}T\times 4\right)$ and the rectangle $\left(12.5-T\right)\times 4$ .<br>cao | *Suvat* equations can be used for two phases of motion. |
| | iii | | EITHER<br>$a=\frac{4}{2.5}=1.6\ \text{m s}^{-2}$<br>$s=\frac{1}{2}\times 1.6t^2=0.8t^2$ | **M1**<br><br>**A1**<br>**[2]** | **1.1a**<br><br>**3.3** | Soi<br><br>FT their $T$ | |
| | | | OR<br>$a=\frac{4}{2.5}=1.6\ \text{m s}^{-2}$<br>$v=\int a\ \text{d}t=1.6t+c$<br>When $t=0,\ v=0$ so $c=0$<br>$s=\int v\ \text{d}t=0.8t^2+c$<br>When $t=0,\ s=0$ so $c=0$<br>Giving $s=0.8t^2$ | **M1**<br><br><br><br><br><br>**A1**<br>**[2]** | | Soi<br>FT their $T$<br><br><br><br><br><br>Must be complete solution – do not award without consideration of +$c$ at least once | |
| | iv | | $0.8t^2=4$<br>$t=\sqrt{5}=2.24\ \text{s}$ | **B1FT**<br>**[1]** | **3.4** | FT their quadratic model in (iii) | |

| Question | | | Answer | Marks | AOs | | Guidance |
|---|---|---|---|---|---|---|---|
| | v | |  | **B1**<br><br>**[1]** | **1.1a** | Must have curved section of the graph decreasing gradient. *S* must be labelled. | |
| | vi | | Total distance (area under the graph) can only be equal if $S > T$<br>If $S > T$<br>  <br>If $S = T$<br>  <br>If $S < T$<br>  | **E1**<br><br>**[1]** | **3.5c** | Needs to give reason relating to the refinement of the model. Graphs not required | "It takes longer to reach 4 $ms^{-1}$" is not sufficient reason |